14.12 Science and Technology

The Madrasas continued to be concerned principally with Muslim theology and its vast literature. In great learning centres like Varanasi, astrology was taught and there was no institution in India, as noted by the French traveller Bernier, to the standards of colleges and universities in Europe. This made the imparting of scientific subjects almost impossible. Attention was, however, given to mathematics and astronomy. Akbar's court poet Faizi translated Bhaskaracharya's famous work on mathematics, *Lilavati*. Despite the presence of Europeans, there was no influence of them on the Indian society during the Mughal period.

The method of water-lift based on pin-drum gearing known as Persian wheel had been introduced during Babur's time. A complicated system of water lift by a series of gear-wheels had been installed in Fatehpur Sikri. Akbar was also credited with popularizing the device of cooling water using saltpetre. He is also the first known person in the world to have devised the 'ship's camel', a barge on which the ship is built to make it easier for the ship to be carried to the sea. Some mechanical devices like the screw for tightening, manually driven belt-drill for cutting diamonds were in use. Agricultural tools continued to be the same, made entirely of wood. In metallurgy, the inability to produce cast iron remained an obvious drawback. As Irfan Habib observed, 'India's backwardness in technology was obvious when the matchlock remained the most common weapon in Indian armies. In Europe the flintlock had long come into use. Indians continued to use the expensive bronze cannon, long after these had become obsolete in Europe. This was because of India's inability to make cast iron even in the seventeenth century.'

Architecture



Architectural progress during the Mughals is a landmark in world art. Mughal buildings were noted for the massive structures decorated with bulbous domes, splendorous minarets, cupolas in the four corners, elaborate designs, and pietra dura (pictorial mosaic work). The mosques built during the time of Babur and Humayun are not of much architectural significance. The Sur dynasty left behind a few spectacular specimens in the form of the Purana Qila at Delhi, and the tombs of Sher Shah and Islam Shah at Sasaram in Bihar. The Purana Qila with a raised citadel and the tombs on a terraced platform surrounded by large tanks were novel features.

During Akbar's reign, Humayun's tomb was enclosed with gardens and placed on a raised platform. Built by Indian artisans and designed by Persian architects it set a pattern to be followed in the future. The Agra fort built with red sandstone is a specimen where Rajput architectural styles were also incorporated. The new capital city of Akbar Fatehpur Sikri enclosed within its walls several inspiring buildings. The magnificent gateway to Fatehpur Sikri, the Buland Darwaza, built by Akbar with red sandstone and marble is considered to be a perfect architectural achievement. The mausoleum of Akbar at Sikandra near Agra started by Akbar and completed by Jahangir includes some Buddhist architectural elements. The tomb of Itimad-ud-daula, father of Nurjahan, built by Jahangir was the first Mughal building built completely with white marble.

Mughal architecture reached its apex during the reign of Shah Jahan. The Taj Mahal is a marble structure on an elevated platform, the bulbous dome in the centre rising on a recessed gateway with four cupolas around the dome and with four free-standing minarets at each of its corners is a monument of universal fame. The Red Fort in Delhi, encompassed by magnificent buildings like Diwan-i Aam, Diwan-i-Khas, Moti Mahal and Hira Mahal reflect the architectural skills of the times of Shah Jahan. The Moti Masjid inside the Agra Fort made exclusively of marble, the Jama Masjid in Delhi, with its lofty gateway, series of domes and tall and slender minarets are the two significant mosques built by Shah

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